

COMP9334 Revision Questions for Week 5A

Question 1

We conducted five independent replications of the discrete event simulation of a queueing system and recorded the response time of the first 20,000 jobs in each replication. You can find these simulation results in 5 separate files `trace*` where $*$ = 1,2, ..., 5.

- (a) Program the transient removal procedure in Law and Kelton, Section 9.5.1 and find what the value of w should be. (A scanned copy of Section 9.5.1 can be found on the course web site.)
- (b) After removing the transient, compute the steady state mean response time obtained from these independent replications. Calculate also the 90% confidence interval.

Question 2

You performed 6 independent replications for a simulation. The mean response times from the 6 replications are: 4.56, 5.23, 5.12, 6.15, 5.57 and 5.34. Answer the following questions.

- (a) Determine the 90% confidence interval.
- (b) Determine the 95% confidence interval.
- (c) Which confidence interval is wider? Why?